

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination
Lab Course CS-305-MJ-P
Operating System-I

Duration: 3 Hours**Maximum Marks:35**

Q.1 Write a C program to illustrate the concept of orphan process. Parent process creates a child and terminates before child has finished its task. So child process becomes orphan process. (Use fork(), sleep(), getpid(), getppid())

[15 Marks]

Q.2 Write a C program that behaves like a shell which displays the command prompt '\$'. It accepts the command, tokenize the command line and execute it by creating the child process. Also implement the additional command 'search' as

- a. \$ search f filename pattern : It will search the first occurrence of pattern in the given file
- b. \$ search a filename pattern : It will search all the occurrence of pattern in the given file
- c. \$ search c filename pattern : It will count the number of occurrence of pattern in the given file

[15 Marks]

Q.3. Oral/Viva

[5 Marks]